Draw a line!

1 MAKE IT!

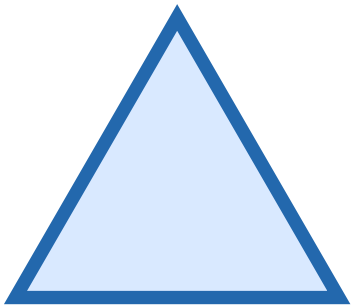
```
my_line = [
    ("up", -3, 0),
    ("heading", 0),
    ("color", "blue"),
    ("width", 4),
    ("down",),
    ("forward", 6),
]
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Blue Line", my_line),
]
```

Copy the code into student_images.py

Triangle!



1 MAKE IT!

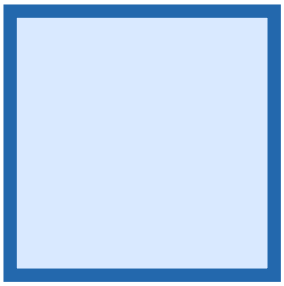
```
my_triangle = triangle(x=-2, y=-2,
                       size=3, angle=45,
                       color="blue", fill="skyblue",
                       speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
# Add triangle to your sequence
image_sequence = [
    ("Triangle", my_triangle),
]
```

Copy the code into student_images.py

Square!



1 MAKE IT!

```
my_square = square(x=-1, y=-1, size=2,
                   angle=20, color="green",
                   fill="lightgreen")
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Square", my_square),
]
```

Copy the code into student_images.py

Rectangle!



1 MAKE IT!

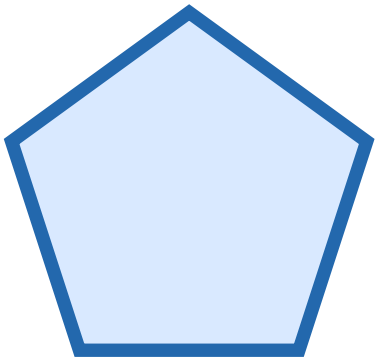
```
my_rectangle = rectangle(x=-3, y=-2,
                         width=5, height=2,
                         angle=0, color="blue",
                         fill="lightblue")
```

2 ADD IT TO YOUR SEQUENCE!

```
# Add rectangle to your sequence
image_sequence = [
    ("Rectangle", my_rectangle),
]
```

Copy the code into student_images.py

Pentagon!



1 MAKE IT!

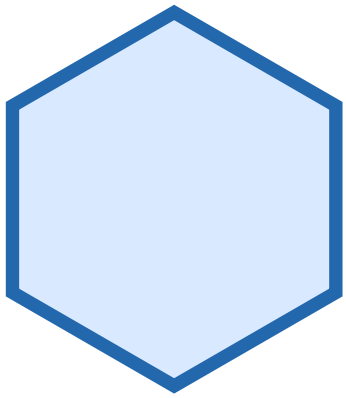
```
my_pentagon = pentagon(x=-1, y=-2,
                       size=2, angle=0,
                       color="orange", fill="yellow")
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Pentagon", my_pentagon),
]
```

Copy the code into student_images.py

Hexagon!



1 MAKE IT!

```
my_hexagon = hexagon(x=-1, y=-2,
                     size=2, color="purple",
                     fill="lavender")
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Hexagon", my_hexagon),
]
```

Copy the code into student_images.py

Star!



1 MAKE IT!

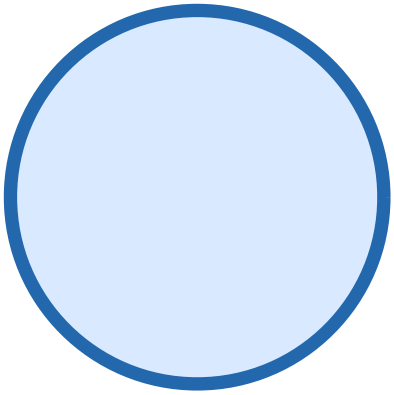
```
my_star = star(x=-2, y=0, size=3,
               angle=0, color="purple",
               fill="yellow", speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Star", my_star),
]
```

Copy the code into student_images.py

Circle!



1 MAKE IT!

```
my_circle = circle(x=0, y=0, radius=1,
                   color="purple", fill="pink",
                   speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Circle", my_circle),
]
```

Copy the code into student_images.py

Spiral!



1 MAKE IT!

```
my_spiral = spiral(x=0, y=0,
                   turn=90, steps=12,
                   growth=0.2, color="purple",
                   speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Spiral", my_spiral),
]
```

Copy the code into student_images.py

Letter a!



1 MAKE IT!

```
my_a = letter_a(x=-3, y=-2, size=3,
                color="red", speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Letter A", my_a),
]
```

Copy the code into student_images.py

Letter h!



1 MAKE IT!

```
my_h = letter_h(x=-2, y=-2, size=3,
                color="purple", speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Letter H", my_h),
]
```

Copy the code into student_images.py

Curved letter s!



1 MAKE IT!

```
my_s = letter_s(x=0, y=2, radius=1,
                color="purple", speed=5)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Letter S", my_s),
]
```

Copy the code into student_images.py

Venetian blinds!



1 MAKE IT!

```
my_blinds = venetian_blind(
    x=-4, y=-3, width=8,
    slats=7, slat_height=0.55,
    gap=0.3, color="steelblue",
    fill="lightblue", speed=5
)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Venetian Blinds", my_blinds),
]
```

Copy the code into student_images.py

Zigzag!



1 MAKE IT!

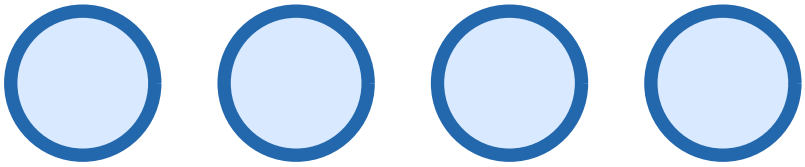
```
my_zigzag = zigzag(x=-4, y=0,
                   steps=6, segment=1,
                   angle=45, color="orange")
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Zigzag", my_zigzag),
]
```

Copy the code into student_images.py

Repeat a shape!



1 MAKE IT!

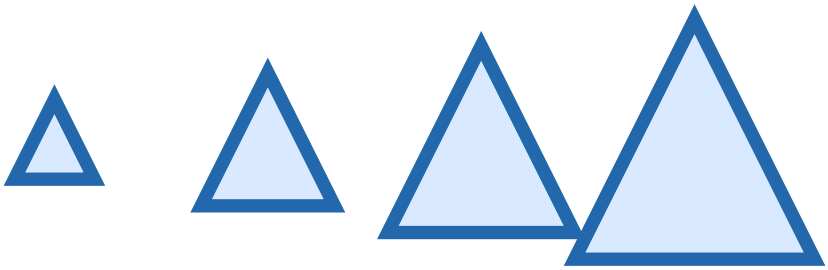
```
my_row = repeat_shape(
    circle, count=4,
    x=-4, y=0, dx=2, dy=0,
    radius=0.4, color="blue"
)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Repeated Shapes", my_row),
]
```

Copy the code into student_images.py

Growing shapes!



1 MAKE IT!

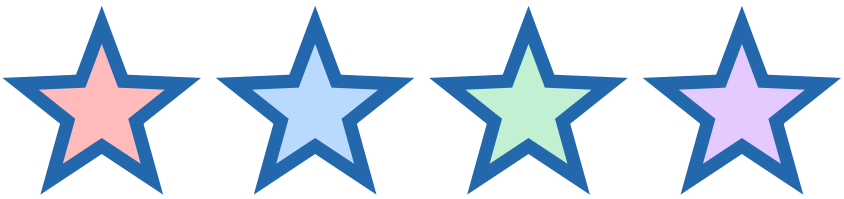
```
my_growth = growing_shapes(
    triangle, count=4,
    x=-3, y=-3,
    start_size=1, step=0.5,
    color="green"
)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Growing Shapes", my_growth),
]
```

Copy the code into student_images.py

Use random colours!



1 MAKE IT!

```
my_colours = random_colors(
    star, count=4,
    x=-4, y=0, dx=2,
    size=1,
    palette=["red", "blue", "green", "purple"]
)
```

2 ADD IT TO YOUR SEQUENCE!

```
image_sequence = [
    ("Random Colours", my_colours),
]
```

Copy the code into student_images.py

Choose a background!

1 MAKE IT!

```
# Put this near the top of student_images.py
background = "beach"
# Other choices:
# "space", "forest", "city", "classroom", "none"
# Or add a PNG/GIF to backgrounds/:
# background = "my_picture.png"
```

2 ADD IT TO YOUR SEQUENCE!

```
# Backgrounds are not added to image_sequence.
sun = circle(x=2, y=2, radius=0.7, fill="yellow")
image_sequence = [("Sun", sun)]
```

Copy the code into student_images.py

Create a scene!!

1 MAKE IT!

```
background = "space"
from turtle_shapes import *
sun = circle(x=2, y=2, radius=0.7,
             color="orange", fill="yellow")
stars = random_colors(
    star, count=3, x=-4, y=1, dx=1.5,
    size=0.7, palette=["yellow", "white"]
)
planet = circle(x=-1, y=-2, radius=1,
                color="purple", fill="pink")
image_sequence = [
    ("Sun", sun),
    ("Stars", stars),
    ("Planet", planet),
]
```

2 ADD IT TO YOUR SEQUENCE!

```
# The scene code above already includes
 # image_sequence.
```

Copy the code into student_images.py

Rotating Square!



1 MAKE IT!

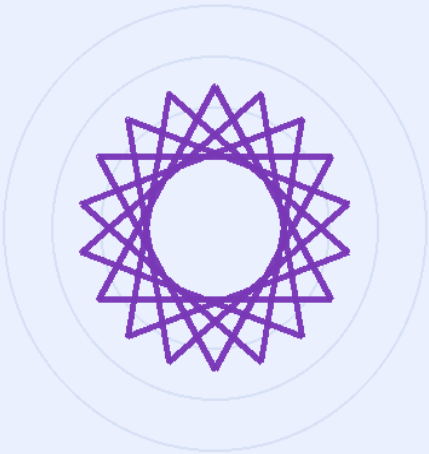
```
rotating_squares = []
for angle in range(0, 360, 30):
    rotating_squares += square(
        x=0, y=0, size=3,
        angle=angle, color="blue")
```

2 ADD IT TO YOUR SEQUENCE!

```
# Add rotating squares to your sequence
image_sequence = [
    ("Rotating Squares", rotating_squares),
]
```

Copy the code into student_images.py

Rotating Triangle!



1 MAKE IT!

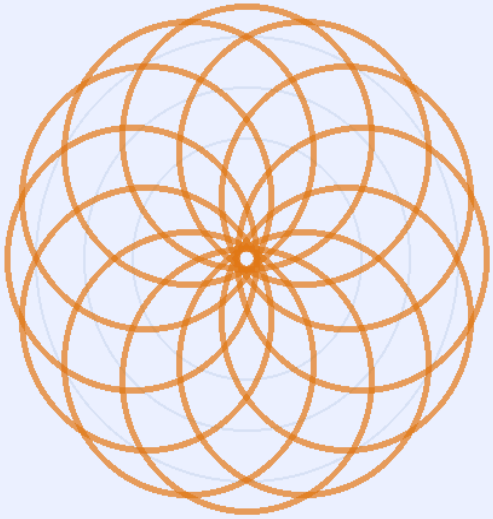
```
rotating_triangles = []
for angle in range(0, 360, 30):
    rotating_triangles += triangle(
        x=0, y=0, size=3,
        angle=angle, color="purple")
```

2 ADD IT TO YOUR SEQUENCE!

```
# Add rotating triangles to your sequence
image_sequence = [
    ("Rotating Triangles", rotating_triangles),
]
```

Copy the code into student_images.py

Rotating Circle!



1 MAKE IT!

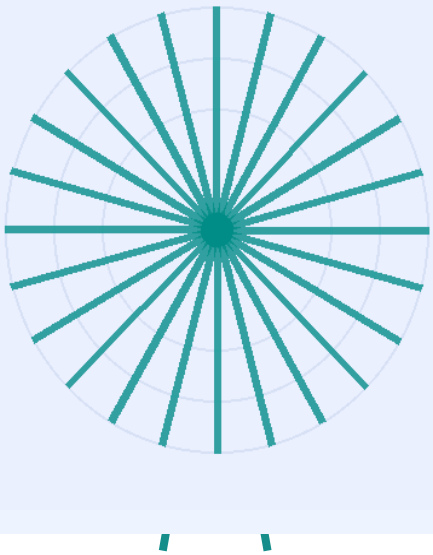
```
from math import cos, sin, radians
rotating_circles = []
for angle in range(0, 360, 45):
    a = radians(angle)
    rotating_circles += circle(
        x=2 * cos(a), y=2 * sin(a),
        radius=1, color="orange")
```

2 ADD IT TO YOUR SEQUENCE!

```
# Add rotating circles to your sequence
image_sequence = [
    ("Rotating Circles", rotating_circles),
]
```

Copy the code into student_images.py

Rotating Line!



1 MAKE IT!

```
rotating_lines = []
for angle in range(0, 360, 20):
    rotating_lines += [
        ("up", 0, 0),
        ("heading", angle),
        ("color", "teal"),
        ("width", 3),
        ("down",),
        ("forward", 3),
    ]
```

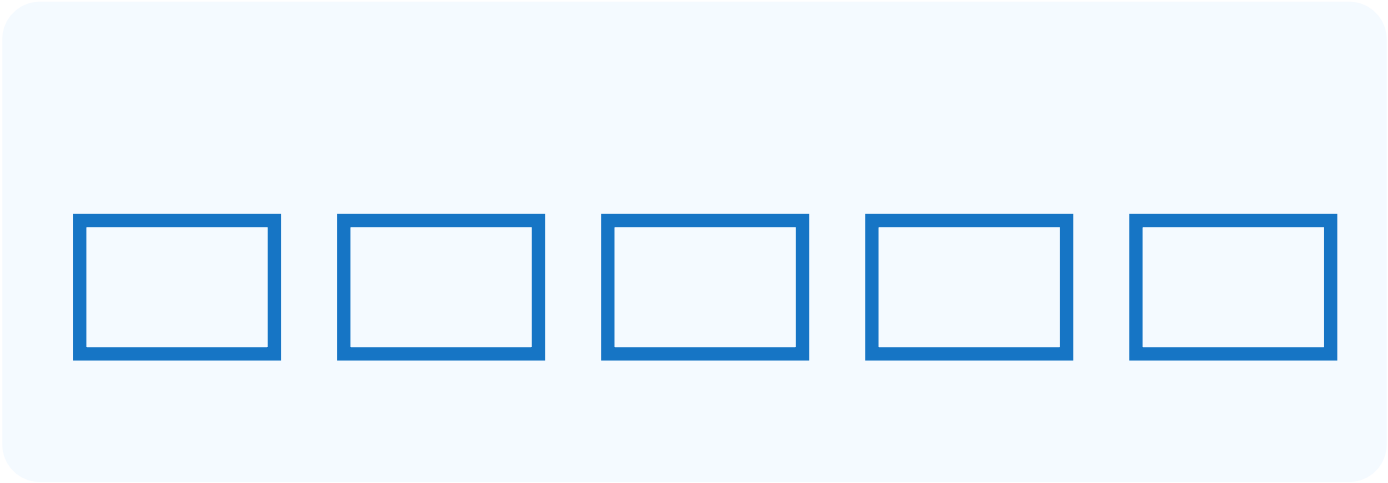
2 ADD IT TO YOUR SEQUENCE!

```
# Add rotating lines to your sequence
image_sequence = [
    ("Rotating Lines", rotating_lines),
]
```

Copy the code into student_images.py

Repeat a Shape!

Make 5 rectangles with one simple command.



1 MAKE IT!

```
# Repeat 5 blue rectangles
my_rectangles = repeat_shape(
    rectangle, count=5, x=-8, y=0,
    spacing=4, width=3, height=2,
    color="blue"
)
```

2 ADD IT TO YOUR SEQUENCE!

```
# Put this inside image_sequence = [ ... ]
("5 Rectangles", my_rectangles),
```

Try changing count=5 to count=3 or spacing=4 to spacing=2!

Repeat Your Own Drawing!

Make a line, then copy it again and again.



1. Make your drawing

```
my_line = [
    ("up", -5, -1),
    ("color", "blue"),
    ("width", 8),
    ("heading", 55),
    ("down",),
    ("forward", 2),
]
```

2. Repeat it 5 times

```
my_lines = repeat_shape(my_line, count=5, spacing=2)
```

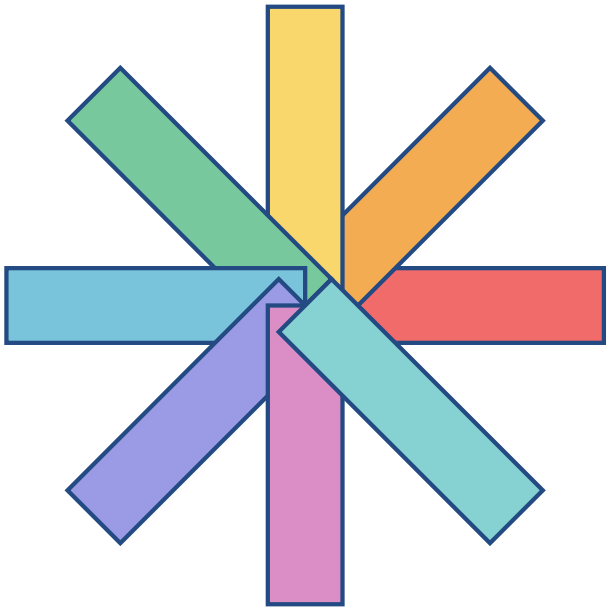
3. Add it to your sequence

```
# Add repeated lines to your sequence
("Repeated Lines", my_lines),
```

Try changing count or spacing to create your own pattern!

Rotate + Fill Repeated Shapes!

Turn each copy, choose one fill colour, or make every copy colourful.



8 rectangles at the same position, rotating 45 degrees each time

1. Rotate + random colours

```
spinning_colours = repeat_shape(
    shape_rectangle, count=8, x=0, y=0,
    dx=0, dy=0, rotate=45,
    width=3, height=1, random_fill=True
)
```

2. One fill colour + vertical movement

```
rising_stars = repeat_shape(
    star, count=4, x=0, y=-5, dx=0, dy=3,
    rotate=20, size=2, fill="yellow"
)
```

3. Add your drawing to image_sequence

```
image_sequence.append(("Spinning Colours", spinning_colours))
image_sequence.append(("Rising Stars", rising_stars))
```

What do the new options mean?

rotate=45: turn each new copy another 45 degrees.

fill="yellow": use yellow for every copy.

random_fill=True: pick a random fill colour for each copy.

palette=["red", "blue"]: limit random fills to those colours.

For custom drawings, rotation uses the first ("up", x, y) as its centre.

Tip: The student_images.py examples are ready to uncomment and try!